BIOSPHERE 2 / RAINFOREST

Virtual Proving Ground for RainForest Climate Strategies

Supporting climate research and student learning

64 scheduled controls **36 climate channels**

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Academic supervision: Dr. Ildar Gabitov

Shared evaluation of climate strategies

Additional scenario testing supports existing research and student projects

Strategies from research groups → Shared simulation scenarios → Results for team discussion

Agreed inputs, documented assumptions, reproducible results

Additional scenario testing alongside each team's own analyses and evidence

Forward models for scenario testing

Dynamics connect a 64-value action to the next 36-channel climate state

STATE

$x(k)$
36 T/RH channels

ACTION

$u(k)$
64 control values

CONTEXT

$w(k)$
weather and mode

$$x(k+1) = F_\theta(x(k), u(k), w(k))$$

Then evaluate climate deviation, energy use, equipment wear, and hard constraints

Each engine documents its assumptions and validation evidence

Support for planning field studies

Offline scenario testing complements operator-led review

26

air handlers support the facility

about 20

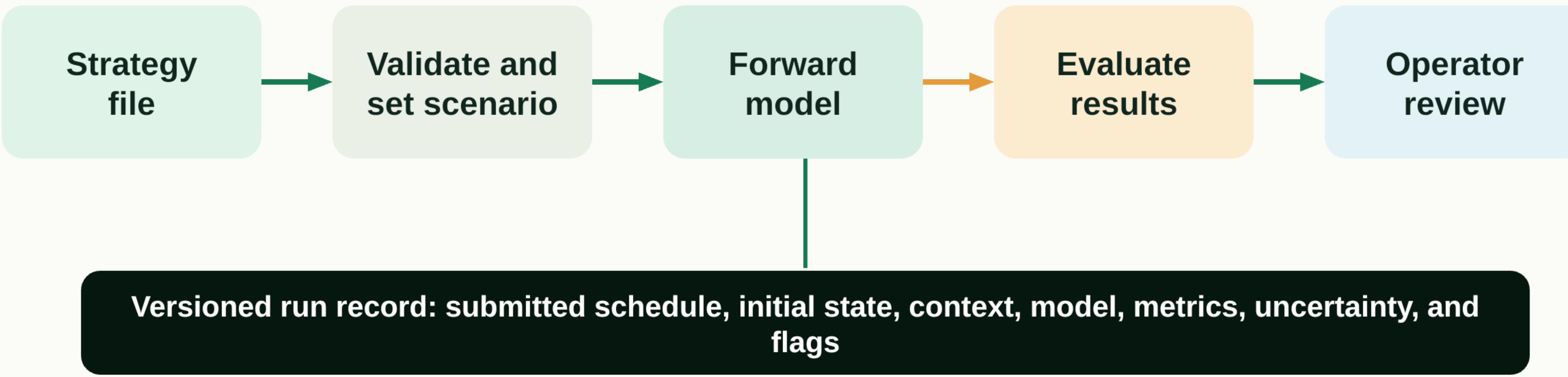
minutes before a sunny-day power loss could irreparably damage biomes

Researchers and operators plan field studies together

Facility context from Biosphere 2. The figures describe environmental support, not algorithm-specific risk.

Virtual Proving Ground architecture

A shared workflow supports strategy authors and Biosphere 2 operations



The platform tests externally created strategies. It does not generate or rewrite them.

Climate strategy contract

A strategy is a timestamped sequence of complete commands

timestamp	complete absolute control vector
08:00	u(1) = [u0, u1, ... , u63]
10:00	u(2) = [u0, u1, ... , u63]
12:00	u(3) = [u0, u1, ... , u63]
14:00	u(4) = [u0, u1, ... , u63]
16:00	u(5) = [u0, u1, ... , u63]

15
setpoints

7
fans

42
valves and dampers

JSON or CSV / units / bounds / monotone timestamps / no missing controls

Illustrative interface demonstration

The current app demonstrates the evaluation workflow, not a validated prediction

35-second interface replay

[Open live demo]

**Illustrative scenario
not yet model-generated**

Reading the simulation output

A useful result explains trajectory, uncertainty, and the reason for each flag

- 36-channel spatial trajectory
- Selected sensor time series
- Comfort and safety envelope
- Energy and stability indicators
- Uncertainty and diagnostic reason

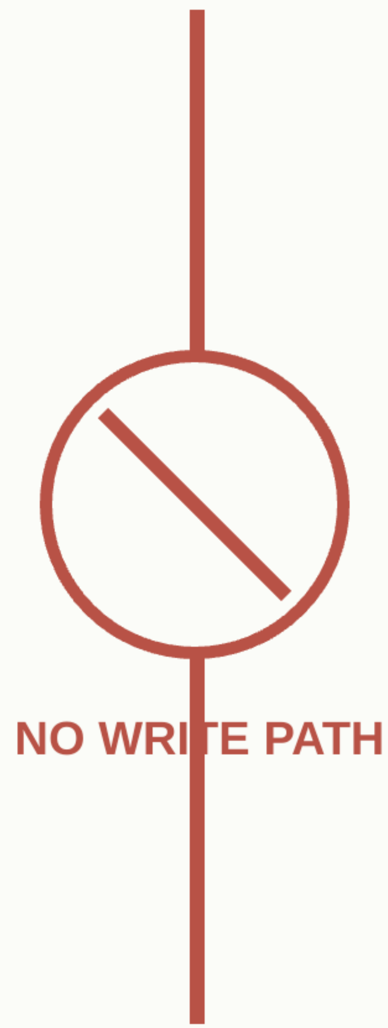
Every warning traces to a channel, timestamp, threshold, and submitted control schedule

Human approval and operational boundaries

The first phase remains offline and advisory

Research environment

- Upload strategy
- Simulate and score
- Preserve run record



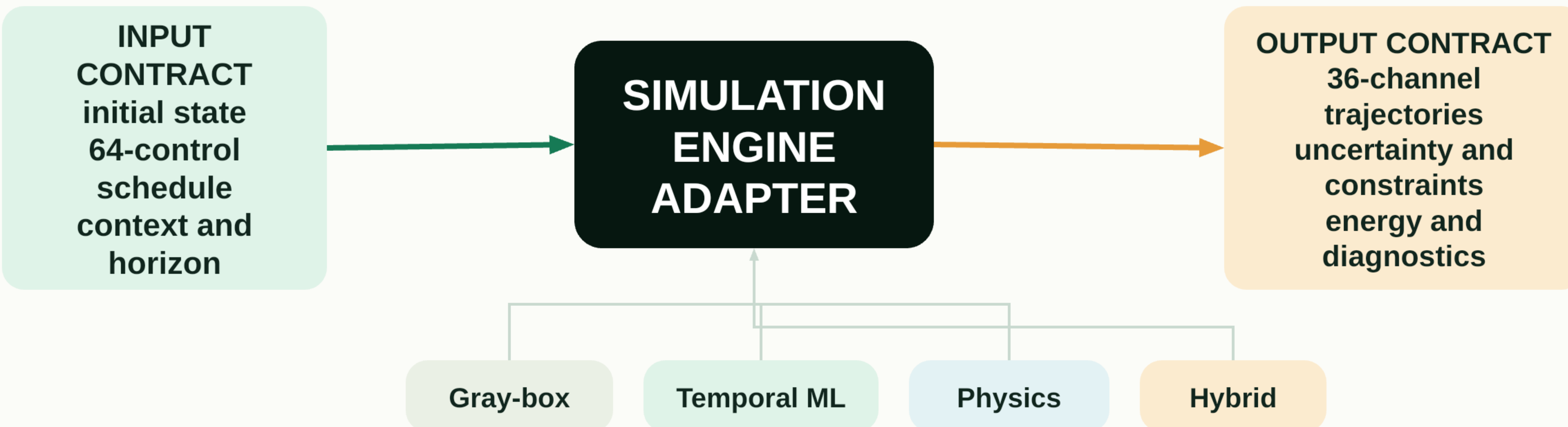
Biosphere 2 operations

- Review evidence
- Approve any future protocol
- Retain stop authority

A prediction supports a decision. It does not grant operational authority.

Pluggable simulation engine

A stable contract allows models to improve without rebuilding the platform



The same strategy and scenario can be compared across engines

Candidate forward-model families

The PoC should benchmark complementary approaches

GRAY-BOX	TEMPORAL ML	PHYSICS	HYBRID
Interpretable dynamics	Learns delays and nonlinear response	Explicit HVAC and energy structure	Physical constraints plus learned residual
Modest data need	Fast repeated simulation	Transparent assumptions	Balances structure and fit
May miss nonlinear coupling	Needs coverage and OOD checks	Calibration can be labor intensive	More integration complexity
simple and legible	**flexible surrogate**	**component fidelity**	**candidate end state**

Select the simplest engine that passes the agreed validation gates

Established simulation foundations

The proposal applies mature testing patterns to a unique biome

1

BOPTEST

controller API
and KPIs

2

EnergyPlus + Modelica

dynamic HVAC
and buildings

3

FMI / FMU

portable model
interfaces

4

ASHRAE 140

systematic model
testing

These references support the engineering pattern. They do not certify the present app or remove RainForest-specific calibration.

Student learning through simulation

Students test their projects and use the results to refine their designs

Student project → **Scenario testing** → **Result interpretation** → **Refined project**

- Practice experimental design and interpret model uncertainty
- Use faculty feedback and versioned runs to improve project designs
- Explore PID, MPC, RL, and other methods in an offline learning environment

Operational and economic value

Scenario comparisons support climate and energy research

Energy-cost-saving potential

requires calibrated evidence before a percentage or dollar claim

Protect required climate conditions → **Compare strategies under the same scenario** → **Explore energy and equipment use**

Operational planning / focused field trials / grant evidence / reusable research benchmarks

Sixteen-week Proof of Concept

The PoC ends with an evidence-based decision on Phase 2

01
WEEKS 1-4
Data synchronization and constraints

02
WEEKS 5-8
Engine benchmark and uncertainty

03
WEEKS 9-12
Module API, scenarios, KPIs, and interface

04
WEEKS 13-16
Historical replay, review, and acceptance

Exit package: offline testbed, validation report, limitations, and Phase 2 recommendation

Decision requested

Approve a scoped offline PoC with operational access and review

Approve the 16-week offline Proof of Concept

- Assign a Biosphere 2 operations liaison
- Authorize synchronized control, climate, weather, energy, and constraint data
- Review acceptance evidence before any Phase 2 or field trial

Live illustrative interface

Dimitri Bolt
Academic supervision: Dr. Ildar Gabitov

Our prototype and planned development

Our current interface is illustrative and our PoC includes engine validation

OUR CURRENT PROTOTYPE

- 64-control inventory
- 36-channel measurement schema
- Data-pipeline code
- Timestamped strategy contract
- Working illustrative 3D interface

OUR PLANNED POC WORK

- Synchronized causal dataset
- Validated and versioned engine
- Uncertainty and unsupported-region checks
- Energy indicators and operating constraints
- Historical replay and operator acceptance

No survey-grade CAD, E57 conversion, equipment write access, or autonomous control in this phase

Required data and validation gates

Our engine evaluation requires aligned histories and protected holdout periods

REQUIRED EVIDENCE

- Commands + actual actuator states
- T/RH + quality flags
- Weather + operating modes
- Energy or equipment power
- Maintenance and schedule events

VALIDATION GATE

ACCEPTANCE CHECKS

- Chronological holdout periods
- Error by channel and horizon
- Constraint-event detection
- Uncertainty calibration
- Reject unsupported scenarios

Data alignment is the first technical risk

Q&A: accuracy and observability

Prediction quality must be reported by use case, horizon, and operating region

01 **How accurate must the simulator be?**

Use-case thresholds. Report error by channel, horizon, season, and operating region.

02 **What if sensors are missing or unreliable?**

Flag degraded operation. Reject the run or widen uncertainty when evidence is insufficient.

03 **How will causal response be validated?**

Align controls, actuator states, context, and protected chronological holdout periods.

Full answers appear in the speaker notes and companion script

Q&A: safety and operational authority

Simulation evidence supports operators and does not bypass them

04 **Could the simulator damage equipment?**

No. The PoC has no write path to equipment.

05 **Who authorizes a field trial?**

Biosphere 2 operations retains authority for every field trial.

06 **How is cybersecurity handled?**

Use offline copies, least privilege, versioning, and network separation.

Full answers appear in the speaker notes and companion script

Q&A: model and platform choices

The contract stays stable while evidence determines the engine

07 **Why not use EnergyPlus alone?**

EnergyPlus still needs RainForest-specific representation, calibration, and validation.

08 **Why consider an LSTM?**

An LSTM can learn delayed nonlinear behavior, but it must pass coverage and uncertainty checks.

09 **How can a new engine be added?**

A common input/output contract and adapters isolate every engine.

Full answers appear in the speaker notes and companion script

Q&A: value and schedule

The PoC measures whether a larger investment is justified

10 **How will energy savings be proven?**

Calibrate energy indicators, then validate them against an approved operational baseline.

11 **What resources does the PoC require?**

Authorized data, an operations liaison, implementation time, and ordinary compute.

12 **Why sixteen weeks?**

Four stages cover data, models, platform integration, and operator-reviewed validation.

Full answers appear in the speaker notes and companion script

Q&A: students, publications, and governance

Participation expands only after benchmark and review rules are stable

13 **How is student code isolated?**

Student code runs offline in isolated environments, never on the operational network.

14 **How are publications, data rights, and IP governed?**

Written University and Biosphere 2 rules govern data, review, authorship, and IP.

15 **Who owns and maintains the platform?**

Phase 2 needs an operational owner, technical maintainer, and release process.

Full answers appear in the speaker notes and companion script

References and live demonstration

Authoritative foundations and project resources

Biosphere 2

biosphere2.org/about/about-biosphere-2

Biosphere 2 energy and water

biosphere2.org/research/user-facility-information/energy-and-water

BOPTEST

ibpsa.github.io/project1-boptest

EnergyPlus

energyplus.net

Modelica Buildings Library

simulationresearch.lbl.gov/modelica

Functional Mock-up Interface

fmi-standard.org

ASHRAE Standard 140

data.ashrae.org/standard140

LIVE ILLUSTRATIVE DEMO

rainforest-virtual-proving-ground.bolt.linkpc.net

Project sources: DigitalTwin control and sensor inventories, data-pipeline code, and current interface source